PROBLEM3:

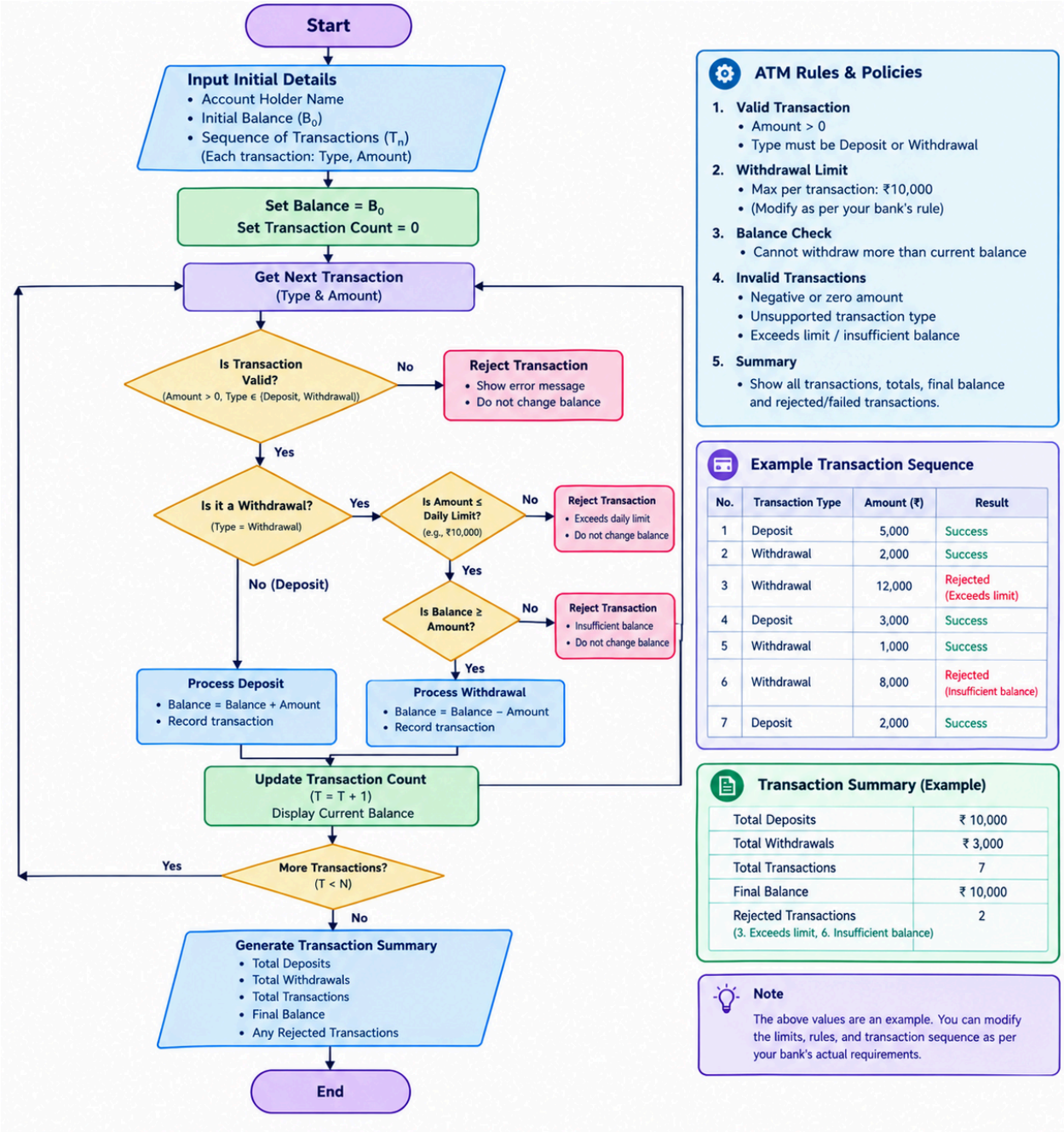
PROBLEM STATEMENT:

. ATM Transaction Sequence Simulation: Process a sequence of deposits and withdrawals, maintain a simulated balance, enforce withdrawal limits, reject invalid transactions, and produce a transaction summary.

ALGORITHM:

1. **Start** the simulation.
2. **Initialize** the starting account balance (e.g., ₹10,000) and the maximum withdrawal limit per transaction (e.g., ₹20,000).
3. **Initialize** tracking variables for the total number of transactions, successful transactions, failed transactions, total deposited amount, and total withdrawn amount.
4. **Display** the initial balance and withdrawal rules to the user.
5. **Repeat** for each transaction in the sequence:
    - Read transaction type (Deposit/Withdrawal) and amount.
    - **Validate Transaction:**
        - Check if the amount is greater than zero. If not, mark as invalid.
        - If withdrawal: Check if the amount exceeds the per-transaction limit or is greater than the current balance. If yes, mark as invalid.
    - **Process Valid Transaction:**
        - If deposit: Add amount to balance, update total deposited.
        - If withdrawal: Subtract amount from balance, update total withdrawn.
        - Record as successful.
    - **Reject Invalid Transaction:**
        - Increment failed transaction count and display error reason.
6. **Print Transaction Summary:** Display final balance, total successful/failed counts, total deposited, and total withdrawn amounts.
7. **End** the simulate.

FLOWCHART:



PSEUDOCODE:

BEGIN ATM_Simulation

```
// Initialize variables
balance = 10000.00
WITHDRAWAL_LIMIT = 20000.00
success_count = 0
fail_count = 0
total_deposited = 0.00
total_withdrawn = 0.00

PRINT "Initial Balance: ₹", balance
PRINT "Per-Transaction Withdrawal Limit: ₹", WITHDRAWAL_LIMIT

// List of transactions: (Type, Amount)
transactions = [
    ("DEPOSIT", 5000),
    ("WITHDRAW", 3000),
    ("WITHDRAW", 25000),  // Exceeds limit
    ("WITHDRAW", 15000),  // Exceeds balance
    ("DEPOSIT", -500),    // Invalid amount
    ("WITHDRAW", 4000)
]

FOR EACH transaction IN transactions DO
    type = transaction.type
    amount = transaction.amount

    PRINT "----------------------------------------"
    PRINT "Processing:", type, "of ₹", amount

    // Validation Checks
    IF amount <= 0 THEN
        PRINT "Error: Invalid amount. Must be greater than zero."
        fail_count = fail_count + 1

    ELSE IF type == "WITHDRAW" THEN
        IF amount > WITHDRAWAL_LIMIT THEN
            PRINT "Error: Exceeds maximum withdrawal limit of ₹", WITHDRAWAL_LIMIT
            fail_count = fail_count + 1
        ELSE IF amount > balance THEN
            PRINT "Error: Insufficient funds. Current balance is ₹", balance
            fail_count = fail_count + 1
        ELSE
            // Process Withdrawal
            balance = balance - amount
            total_withdrawn = total_withdrawn + amount
```

```
            success_count = success_count + 1
            PRINT "Success! New Balance: ₹", balance
        END IF

    ELSE IF type == "DEPOSIT" THEN
        // Process Deposit
        balance = balance + amount
        total_deposited = total_deposited + amount
        success_count = success_count + 1
        PRINT "Success! New Balance: ₹", balance

    ELSE
        PRINT "Error: Unknown transaction type."
        fail_count = fail_count + 1
    END IF
  END FOR

  // Print Summary
  PRINT "========================================"
  PRINT "          TRANSACTION SUMMARY          "
  PRINT "========================================"
  PRINT "Final Balance          : ₹", balance
  PRINT "Successful Transactions: ", success_count
  PRINT "Rejected Transactions  : ", fail_count
  PRINT "Total Deposited        : ₹", total_deposited
  PRINT "Total Withdrawn        : ₹", total_withdrawn
  PRINT "========================================"
END ATM_Simulation
```

PROGRAM:

```
def atm_simulation():
    # Initial setup
    balance = 10000.00
    WITHDRAWAL_LIMIT = 20000.00

    success_count = 0
    fail_count = 0
    total_deposited = 0.00
    total_withdrawn = 0.00

    print("=== ATM Transaction Sequence Simulation ===")
    print(f"Initial Balance: ₹{balance:,.2f}")
```

```
print(f"Per-Transaction Withdrawal Limit: ₹{WITHDRAWAL_LIMIT:,.2f}\n")

# Pre-defined sequence of transactions: (type, amount)
transaction_sequence = [
    ("DEPOSIT", 5000),
    ("WITHDRAW", 3000),
    ("WITHDRAW", 25000),  # Violation: Exceeds limit
    ("WITHDRAW", 15000),  # Violation: Insufficient balance
    ("DEPOSIT", -500),    # Violation: Negative/Zero amount
    ("WITHDRAW", 4000)
]

for i, (t_type, amount) in enumerate(transaction_sequence, 1):
    print(f"Transaction {i}: {t_type} of ₹{amount:,.2f}")

    # Validation checks
    if amount <= 0:
        print("❌ Rejected: Amount must be greater than zero.")
        fail_count += 1

    elif t_type == "WITHDRAW":
        if amount > WITHDRAWAL_LIMIT:
            print(f"❌ Rejected: Exceeds single-transaction limit of ₹{WITHDRAWAL_LIMIT:,.2f}.")
            fail_count += 1
        elif amount > balance:
            print(f"❌ Rejected: Insufficient funds. Current balance is ₹{balance:,.2f}.")
            fail_count += 1
        else:
            balance -= amount
            total_withdrawn += amount
            success_count += 1
            print(f"✅ Approved. Current Balance: ₹{balance:,.2f}")

    elif t_type == "DEPOSIT":
        balance += amount
        total_deposited += amount
        success_count += 1
        print(f"✅ Approved. Current Balance: ₹{balance:,.2f}")
    else:
        print("❌ Rejected: Invalid transaction type.")
        fail_count += 1

    print("-" * 45)
```

```
    # Summary Report
    print("\n==========================================")
    print("          TRANSACTION SUMMARY           ")
    print("==========================================")
    print(f"Final Account Balance   : ₹{balance:,.2f}")
    print(f"Successful Transactions : {success_count}")
    print(f"Rejected Transactions   : {fail_count}")
    print(f"Total Amount Deposited  : ₹{total_deposited:,.2f}")
    print(f"Total Amount Withdrawn  : ₹{total_withdrawn:,.2f}")
    print("==========================================")

if __name__ == "__main__":
    atm_simulation()
```

OUTPUT:

```
=== ATM Transaction Sequence Simulation ===
Initial Balance: ₹10,000.00
Per-Transaction Withdrawal Limit: ₹20,000.00

Transaction 1: DEPOSIT of ₹5,000.00
✅ Approved. Current Balance: ₹15,000.00
Transaction 2: WITHDRAW of ₹3,000.00
✅ Approved. Current Balance: ₹12,000.00
Transaction 3: WITHDRAW of ₹25,000.00
❌ Rejected: Exceeds single-transaction limit of ₹20,000.00.
Transaction 4: WITHDRAW of ₹15,000.00
❌ Rejected: Insufficient funds. Current balance is ₹12,000.00.
Transaction 5: DEPOSIT of -₹500.00
❌ Rejected: Amount must be greater than zero.
Transaction 6: WITHDRAW of ₹4,000.00
✅ Approved. Current Balance: ₹8,000.00
          TRANSACTION SUMMARY
Final Account Balance   : ₹8,000.00
Successful Transactions : 3
Rejected Transactions   : 3
Total Amount Deposited  : ₹5,000.00
Total Amount Withdrawn  : ₹7,000.00
```